CHAPTER 7

Sensors

Book: Fundamentals of Mechatronics by Jouaneh

CHAPTER OBJECTIVES:

When you have finished this chapter, you should be able to:

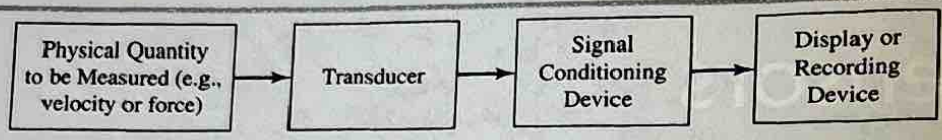
- Interpret a sensor performance specification
- Select a specific sensor for a given measurement application
- Explain the different types of displacement, proximity, speed, temperature, and vibration sensors
- Predict the output of strain gage-based sensors
- Explain the principle of operation of many of the different sensors covered in this chapter
- Explain the use of filters in signal processing
- Analyze a bridge circuit to process the output of resistance-type sensors

7.1 INTRODUCTION

Sensors are vital components of mechatronic systems, since they provide information that allows us to monitor and to control the operation of these systems. Without the availability of sensory information, automated systems cannot operate. A sensor is an element that produces an output in response to changes in physical quantity (such as temperature, force, or displacement). The active element of some sensors is called a **transducer**, which is the part of the sensor that converts the physical quantity (such as the force or displacement into an equivalent electrical signal in the form of voltage or current). The physical quantity changes a property of the transducer (such as its resistance, inductance, or magnetic coupling). Through electronic circuits, these property changes of the transducer are converted into a low-level voltage or current electrical signals. The terms sensor and transducer are sometimes treated as synonyms, but note that not all sensors produce an electrical signal as an output. Examples include the mercury bulb thermometer and the spring scale force sensor. Figure 7.1 shows a block diagram of the process of measurement using a sensor with a transducer. Normally, the output from the transducer is not in a form suitable to be read by a display device or meter, and signal conditioning operations (such as filtering or amplification) are needed to process the output.

There are a variety of sensors available that are commonly used. These include sensors that measure motion-related information (such as strain, speed, displacement, and acceleration). Also, sensors are available to measure process parameters (such as temperature, level, and pressure). This chapter will focus more on sensors that

Figure 7.1
Measurement process



measure motion-related information. The next chapter will discuss electric actuators. Both sensors and actuators are key to implementing feedback control of motion-control systems. We start this chapter by discussing some of the performance parameters of sensors. For further reading on sensors and measurements, see [27-29].

7.2 Sensor Performance Terminology

There are a number of parameters that characterize sensors' performance. The time-independent characteristics are called the static characteristics, while the time-dependent characteristics are called the dynamic characteristics. The **static characteristics** characterize the sensor output after it has settled due to changes in the physical quantity being measured. The **dynamic characteristics** describe the sensor characteristics from the time the physical quantity has changed to the time before the output has settled.

7.2.1 Static Characteristics

Range Minimum to maximum value that can be measured is the range. The range defines the allowable range of the physical quantity that can be detected by the sensor.

Accuracy The difference between true and actual measured value is the accuracy. It is commonly expressed as a percentage of full-scale value. For example, if a temperature sensor has a range of 0 to 200°C and an accuracy of +/−0.5% full-scale value, then the temperature read by the sensor is off from the true actual temperature by +/−1°. Note that the accuracy error can be improved by calibration.

Sensitivity The relationship between the measured input and the output of the sensor is its sensitivity. If the sensor has a linear input–output relationship, then the sensitivity is the slope of this curve. Sometimes, this parameter is used to indicate the sensitivity of the sensor to non-measured input (response due to transverse motion when the sensor is designed to measure axial motion) or the environment (temperature).

Resolution The smallest change in input value that will produce an observable change in the output is the resolution. The inherent resolution should be distinguished from the display device resolution.

Hysteresis The maximum difference in sensor output for the same input quantity is the hysteresis, with one measurement taken while the input was increasing from zero and the other by decreasing the input from the maximum input. A sensor with hysteresis will have a different output value that is a function of whether the input quantity was increasing or decreasing when the measurement was made. The hysteresis error is illustrated in Figure 7.2.

Repeatability Error in output value for repeated application of the same input value is called repeatability or precision. The smaller the repeatability error, the higher the measurement precision would be. Repeatability is affected by signal

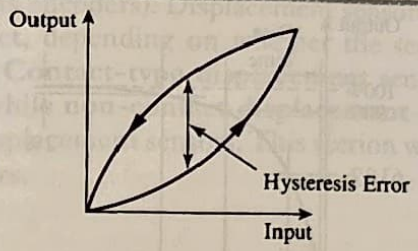
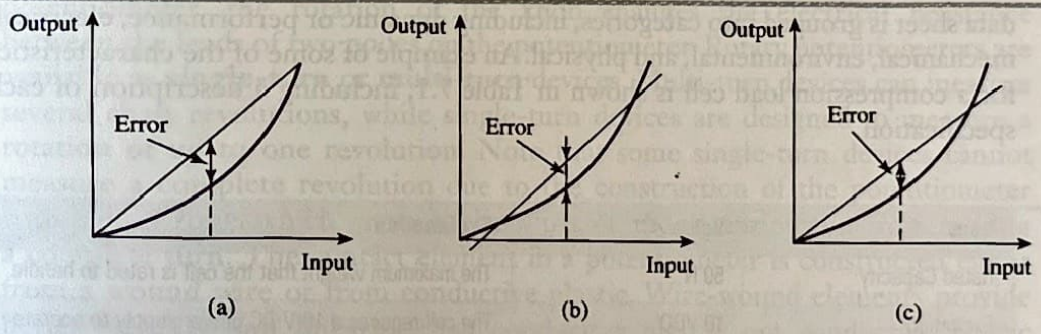


Figure 7.2
Illustration of hysteresis error

interference, vibration, and temperature fluctuation. Repeatability error cannot be reduced by calibration.

Non-Linearity Error Most sensors are designed to have a linear output, but their output is not perfectly linear. The non-linearity error is a measure of the maximum difference between the sensor actual output and a straight line fit to the sensor input–output data and is usually specified as a percentage of the full-scale output. There is no unique way to obtain the straight line fit. The straight line can connect the minimum and maximum output values that define the sensor range, or it can be obtained from a least-square fit to the entire input–output data or from a least-square fit to the input–output data with one end of the line passing through the origin. Figure 7.3 illustrates these cases and shows that the magnitude of this error is dependent on how this error is defined.



(a) Minimum to maximum fit
(b) Least-square fit through data
(c) Least-square fit with one end through the origin

Figure 7.3
Illustration of non-linearity error

Stability Stability or drift refers to the variation of the output with time when the input quantity is not changing. When no input is applied to the sensor, the output variation is called the zero drift. Stability affects the repeatability of the measurement.

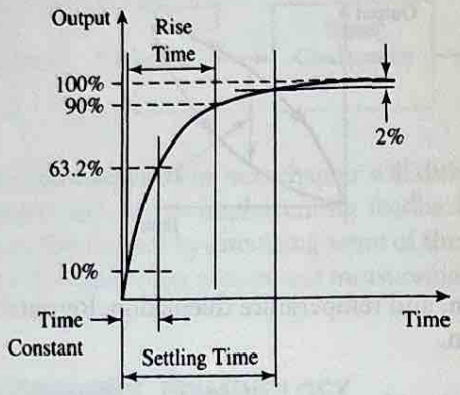
7.2.2 Dynamic Characteristics

Rise Time The time it takes the output to change a certain percentage is the rise time. A common measure is the time for the output to change from 10 to 90% of the final steady–state value.

Time Constant This is defined as the time it takes the output to reach 63.2% of the final output. A large time constant implies a sluggish sensor, while one with a small value indicates a rapidly responding sensor. If the sensor has a first-order response characteristics, then it takes about four time constants to reach the final value when subjected to a step input.

Settling Time The time it takes the output to reach within certain percentage of the final steady–state value is the settling time. A common value is the 2%

Figure 7.4
Illustration of basic dynamic response characteristics



settling time. The rise time, time constant, and settling time are illustrated in Figure 7.4 for a sensor with first-order response characteristics.

Bandwidth The bandwidth defines the frequency range for which the sensor is designed to operate. At the bandwidth frequency, the sensor output will be 70.7% of the DC level. The sensor can operate at frequencies higher than the bandwidth, but the output of the sensor will be significantly diminished. When a sensor is used to provide feedback information in a closed loop control system, the sensor bandwidth should be larger than the controller bandwidth.

Values for each of these sensor performance characteristics are found in the manufacturer data sheet for the particular sensor. Normally, the specification in the data sheet is grouped into categories, including dynamic or performance, electrical, mechanical, environmental, and physical. An example of some of the characteristics for a compression load cell is shown in Table 7.1, including a description of each specification.

Table 7.1
An example of the specifications for a load cell sensor

Item	Value	Explanation
Rated Capacity	50 N	The maximum weight that the cell is rated to handle
Excitation	10 VDC	The cell requires a 10-V DC power supply to operate
Rated Output	2 mV/V nominal	The nominal cell output at the maximum load (10 lbs) will be 20 mV (2 mV/V × 10 V)
Linearity	+/−0.25% FS	With a 25-N load applied, the cell output will indicate a load of 25 N +/−0.125 N
Hysteresis	+/−0.15% FS	Due to hysteresis, the cell output can be vary by +/−0.075 N
Maximum Load (Safe Overload)	150% of rated capacity	The allowable increase in the rated capacity. For this load cell, the maximum load should not exceed 75 N
Bridge Resistance	350 Ω	The resistance of the strain gage inside the load cell

7.3 Displacement Measurement

Displacement sensors are ones that provide information about the change in the position of a rigid body. The sensors can be classified as those that provide **analog output** (such as potentiometers and resolvers) and those that provide

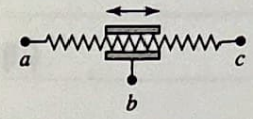
digital output (such as encoders). Displacement sensors also can be classified as contact or non-contact, depending on whether the sensor contacts the object during measurement. **Contact-type displacement** sensors include strain gages and potentiometers, while **non-contact displacement** sensors include encoders and capacitive-type displacement sensors. This section will discuss different types of displacement sensors.

7.3.1 Potentiometers

A potentiometer is a contact-type sensor that provides displacement information by measuring the voltage drop across a resistor. Potentiometers can be of the linear or rotary type. A **linear potentiometer** is designed to measure linear displacement. The sensor has a linear slider that is attached to the object whose displacement needs to be measured (see Figure 7.5). The displacement of the slider changes the electrical resistance between nodes *a* and *b*, which then can be used as a measure of displacement. In normal operation, a DC voltage is applied between nodes *a* and *c*, and the voltage output between nodes *a* and *b* is used as a measure of displacement. If node *a* is connected to the ground, then the *b* node voltage will increase as the slider travels from *a* to *c*. The *b* node is commonly called the wiper.

Figure 7.5

Model of a linear potentiometer



A **rotary-type potentiometer** (see Figure 7.6) is designed to measure angular displacement. The sensor has a rotary knob that is coupled to the shaft of the object whose angular displacement needs to be measured. Similar to a linear potentiometer, the rotation of the knob changes the electrical resistance between the leads of two nodes on the potentiometer. Rotary potentiometers are available as **single-turn** or **multi-turn** devices. Multi-turn devices can measure several shaft revolutions, while single-turn devices are designed to measure a rotation of up to one revolution. Note that some single-turn devices cannot measure a complete revolution due to the construction of the potentiometer with a dead zone, which prevents the wiper on the potentiometer from making a complete turn. The contact element in a potentiometer is constructed either from a wound wire or from conductive plastic. Wire-wound elements provide better stability and linearity than conductive plastic, but conductive plastic offers better resolution and longer life. The use of a wound wire results in a step change in the output voltage (and hence a coarser resolution than conductive plastic) as the slider moves from one turn in the coil to the next turn. Potentiometers have the advantage that they are easy to use, but because they are contact-type devices, they have a frictional resistance, which affects the

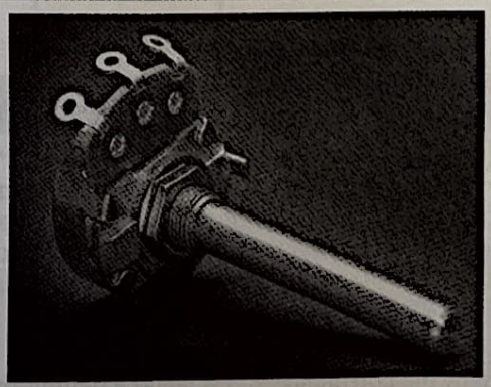
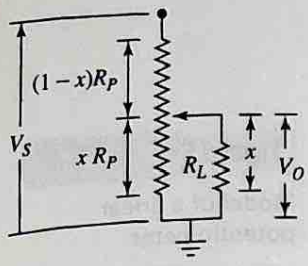


Figure 7.6

A commercial rotary potentiometer
(© Wayne Higgins/Alamy)

Figure 7.7

Model of a potentiometer interfaced with a measuring device with load resistance (R_L)



motion of the measured object. Because potentiometers provide an analog voltage as their output, they also need to be interfaced with an A/D converter before the signal is read by a PC or a microcontroller.

The resistance of a potentiometer is important. A high resistance results in a smaller current and hence less heat loss through the potentiometer while it is in operation, but it also worsens the loading error, since in practice, the output voltage of the potentiometer is read by a device that does not have infinite impedance (see Section 2.6). Loading introduces **nonlinearities** into the potentiometer output. To see this, refer to Figure 7.7, and assume that the potentiometer has a resistance R_P, the measuring device has a resistance R_L, and the supply voltage is V_S. Then the voltage output at any position x ($0 \le x \le 1$) is given by the relationship

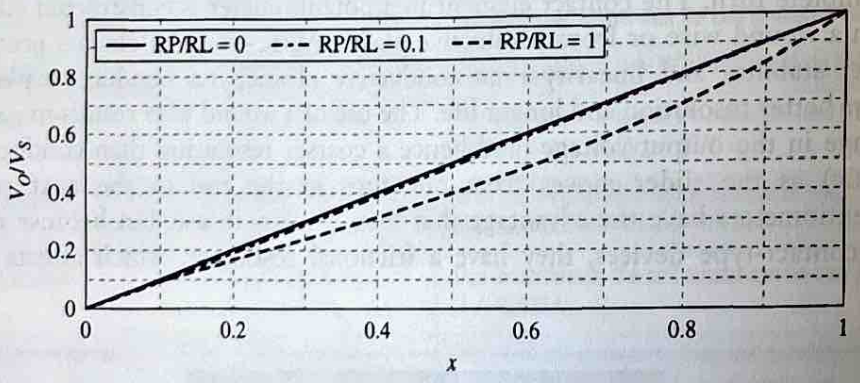
$$V_0 = \frac{xV_S}{1 + x(1 - x)\frac{R_P}{R_L}} \tag{7.1}$$

This relationship is obtained by using the voltage dividing rule to compute the output voltage (V_O) and noting that the load resistor (R_L) and the potentiometer resistor ($x\,R_P$) are two resistors in parallel. Note that if the load resistance (R_L) is infinite, then R_P/R_L is zero, and Equation (7.1) reduces to $V_O = x\,V_S$, where the output voltage is directly proportional to the slider position x. If R_L is not infinite, then the output voltage varies nonlinearly with the slider position x. The nonlinearity worsens as the ratio of R_P/R_L increases.

Figure 7.8 shows a plot of Equation (7.1) for three values of the ratio R_P/R_L: one for $R_P/R_L = 0$, another for $R_P/R_L = 0.1$, and the third for $R_P/R_L = 1.0$. Note how the output voltage varies nonlinearly with x, especially for $R_P/R_L = 1$. To eliminate this problem, one should select a potentiometer with as small a resistance as possible. Example 7.1 illustrates computations for a potentiometer.

Figure 7.8

A plot of Equation (7.1)



Example 7.1 Potentiometer

A single-turn rotary potentiometer with a 330° measurement range is used to provide angular-position feedback information for a positioning application. A 5-V DC voltage is applied across the potentiometer leads, and the potentiometer output is connected to a 12-bit A/D convertor with a +/−5 V range. The potentiometer resistance is 50 Ω. Determine:

a. The effective resolution of this sensor
b. The power loss by the potentiometer, assuming a half-motion displacement

Solution:

a. Assuming that the potentiometer uses a film as the resistance element, then the resolution is determined by the A/D resolution. The 330° motion range is mapped into the 5-V output, and the A/D total voltage range of 10 V is mapped into 2^{12} positions. Thus, the angular resolution is

$$1/4096 \times 10/5 \times 330° = 0.161°$$

b. If the load impedance is considered to be infinite, then all of the current is passed through the potentiometer. In this case, the current is

$$5\ V/50\ \Omega = 0.1\ A$$

and the power loss is i^2R or 0.5 W.

If the load impedance is not infinite, then some of the current will pass through the load, and the power dissipated by the potentiometer will be smaller than computed above.

7.3.2 LVDT

The linear variable differential transformer (LVDT) is a device for measuring mechanical displacement. The device has a simple construction and consists of a moveable iron core surrounded by three transformer coils (see Figure 7.9). An external AC voltage is applied to the center (primary) coil, and the output signal is obtained from the two end (secondary) coils which are connected in opposite phase. The excitation frequency is typically several kHz, but it could range from 100 Hz to 20 kHz. Similar to a regular transformer, the voltage in the secondary coil is proportional to the number of turns in the secondary coil that are coupled to the turns in the primary coil. The position of the core affects the coupling between the center and the two end coils, and thus, the AC output signal is proportional to the displacement of the core relative the windings. The AC output signal, which has the same frequency as the excitation frequency, is processed to produce a DC output signal proportional to the displacement that can be read by an A/D convertor or connected to a display device. Processing the AC signal includes rectification to produce a DC signal, filtering to remove high-frequency signals, and amplification to produce a suitable voltage output level. A low-pass filter with a cut-off frequency about 10% of the excitation frequency is used. The filter passes the components associated with the low-frequency mechanical motion but filters the high-frequency excitation signal.

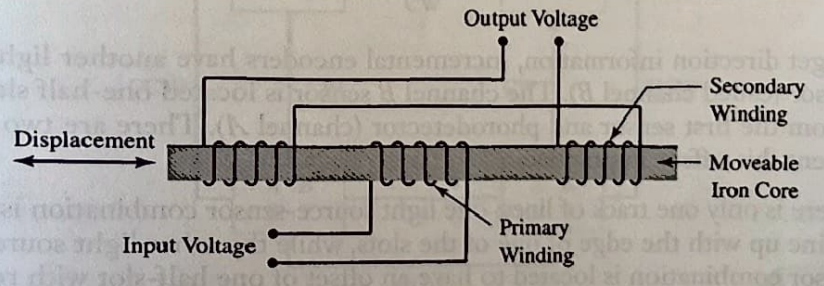


Figure 7.9
LVDT construction

To reduce noise sensitivity, the voltage-output signals from the two secondary coils (A and B) are computed using a **ratiometric formula** where the difference between the voltage signals is divided by the sum of the two signals as

$$V_O = \frac{V_A - V_B}{V_A + V_B} \tag{7.2}$$

An LVDT has the advantage that it can be constructed to measure displacement ranging from a few to several centimeters with almost infinite resolution. There is also no damage from overloading, as overloading simply separates the core from the device, and it is relatively insensitive to temperature changes. However, being a contact displacement sensor, the LVDT has a limited frequency response. Also, signal conditioning is required to process the output signal. LVDTs are available as DC or AC power operated. The DC configuration offers ease of installation and the ability to use battery power in measurement situations where AC power is not available, while the AC version results in a smaller body size and more accurate signal.

7.3.3 Incremental Encoder

Encoders are non-contact, optical-based digital devices that are used for measuring displacement. Similar to potentiometers, they can be used to measure both linear and rotary displacement. We will concentrate on rotary-type encoders, which are widely used in motion-control applications. Rotary encoders are available as incremental or absolute type. An **incremental encoder** measures changes in rotation from some datum position, while an **absolute encoder** measures the actual angular position. When an incremental encoder is used, the motion system needs to be 'homed' to establish reference information.

In its basic form, an incremental encoder is constructed from two light sources that shine light through a disk that has an alternating pattern of black and clear stripes. The light is sensed by two photodetector sensors that are located on the other side of the disk. To understand the operation of an incremental encoder, let us assume first that we have only one light source and one sensor. When the disk rotates, the light signal (as measured by the photo detector sensor) looks like that shown in Figure 7.10. The output of the sensor will be a train of pulses with each pulse corresponding to the light pattern that is captured by the optical sensor while one strip on the disk passes through the light sensor zone. To get angular displacement information, we simply count the number of pulses generated as the disk rotates from one position to another. Unfortunately, this simple scheme cannot provide us with direction information.

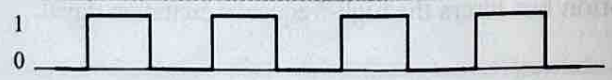


Figure 7.10
Output from a single light/sensor combination

To get direction information, incremental encoders have another light source and sensor (called channel *B*). The channel *B* sensor is located one-half slot-width apart from the first sensor and photodetector (channel *A*). There are two ways to implement this **offset** in practice.

1. There is only one track of lines: one light source-sensor combination is located to line up with the edge of one of the slots, while the other light source-sensor combination is located to have an offset of one half-slot with respect to the edge of one of the slots.
2. Two concentric tracks of lines are used: the slots in one track have an offset of one-half slot with respect to the slots in the other track, but the sensors have no offset between them.

The pattern of the two sensor signals generated for clockwise (CW) and counterclockwise (CCW) rotations of a disk under constant angular speed are shown in

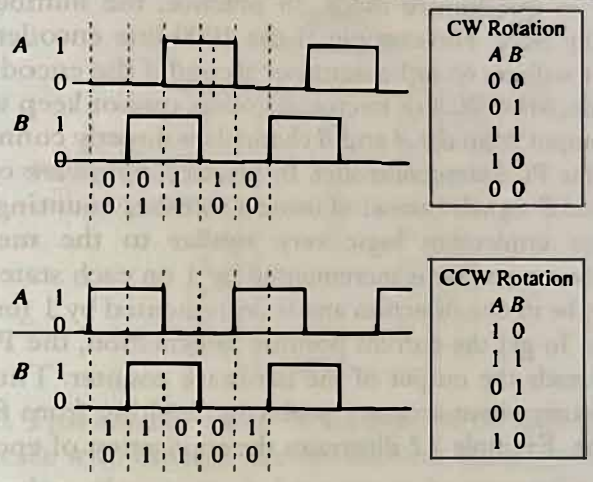


Figure 7.11
Output of an incremental encoder

Note:

Note: Channels A and B, as well as their order, may vary across different books.

Figure 7.11. Notice how the *B* signal leads the channel *A* signal by a quarter of a cycle for CW rotation, and how it lags behind the channel *A* signal by the same amount for CCW rotation. The cycle that we are referring to is the chord distance made up of one black strip and one clear strip. This lead/lag between the two channel outputs enables one to determine the direction of the rotation of the shaft that is attached to the encoder disk.

To understand this further, examine the *A* and *B* channel signal patterns for CW and CCW rotations that are shown in Figure 7.11. Notice that the output switches between one of four possible states for either rotation direction, and the order of these states is different for each direction. For example, if the photodetector sensors output is 00, then the next state will be 10 for CCW rotation and 01 for CW rotation. The different transitions for CW and CCW rotations enable one to write state-transition logic to determine the direction of rotation. Figure 7.12 shows an example of a state-transition diagram (see Section 6.4) that can do this job. Notice that, regardless of what state the output of the two sensors starts at, the diagram can determine the direction of rotation by examining the transitions from any one of the four possible states.

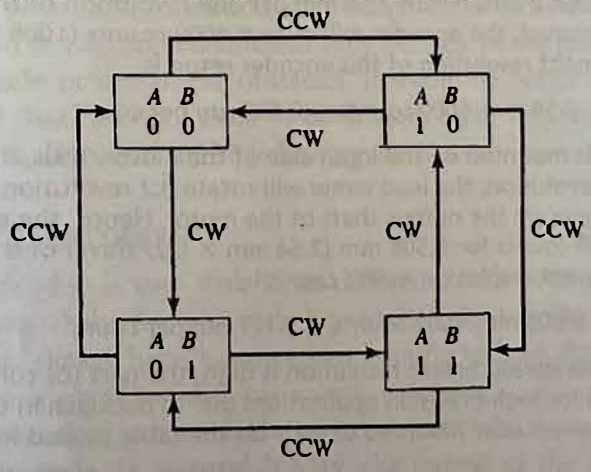


Figure 7.12
State-transition diagram for an incremental encoder

As noted before with the use of two sensors, we get four distinct states for each strip on the disk. Thus, if an encoder has 1000 strips (or lines), we will get 4000 distinct states per one revolution of the encoder disk. Thus, the use of two sensors improves the resolution of the encoder by a factor of 4, since we can count 4000 leading and trailing edges per one revolution compared to counting 1000 pulses per revolution for a single sensor. An encoder that gives four times the number of

lines is operating in **quadrature** mode. In practice, the number of counts per second can get very large. For example, if the 1000-line encoder was rotating at 1000 rpm, then we will get 66.6e3 counts per second if the encoder was operating in quadrature mode. Most PC's or microcontrollers cannot keep up with counting at this rate if the output from the *A* and *B* channels is directly connected to the digital input port of the PC/microcontroller. In practice, hardware counters are used to process the *A* and *B* signals instead of using a software counting solution. These dedicated counters implement logic very similar to the method shown in Figure 7.12. The counter value is incremented by 1 on each state transition if the motion happens to be in one direction and is decremented by 1 for a motion in the opposite direction. To get the current position information, the PC or the microcontroller simply reads the output of the hardware counter. Thus, the processor does not have to worry about accuracy problems resulting from failing to count a particular transition. Example 7.2 illustrates the application of encoders.

Example 7.2 Incremental Encoder

A DC motor equipped with an incremental optical encoder is used to drive a lead-screw positioning table, as shown in Figure 7.13. The screw has a lead of 2.54 mm/rev., the encoder disk has 1000 lines, and the encoder is operated in quadrature mode. Determine the measurement resolution of this encoder for the following.

a. The setup shown Figure 7.13.
b. The motor replaced with a geared one with a 5:1 gear ratio.

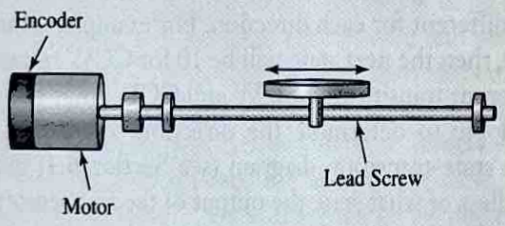


Figure 7.13

Solution:

a. The table travels a distance of 2.54 mm per one revolution of the motor. During this interval, the encoder will output 4000 counts (1000 × 4). Thus, the measurement resolution of this encoder setup is

$$2.54 \text{ mm}/4000 \text{ counts} = 0.635 \ \mu\text{m per count}$$

b. The encoder is mounted on the input side of the motor. Thus, if the motor rotates one revolution, the lead screw will rotate 0.2 revolution due to the use of a 5:1 gear on the output shaft of the motor. Hence, the encoder will generate 4000 counts for 0.508 mm (2.54 mm × 0.2) travel of the table, and the measurement resolution in this case is

$$0.508 \text{ mm}/4000 \text{ counts} = 0.127 \ \mu\text{m per count}$$

Note that while the measurement resolution is high, the part (b) configuration is not normally used for high-precision applications due to backlash in the gears and lead screw. A linear encoder mounted directly on the table is used instead.

Figure 7.14 shows an example of a **commercial counter IC**. The LS7166 IC is a 24-bit counter that can count in different modes, including up/down and quadrature. From Example 7.2 using 1000 rpm rotation speed and a 1000-line encoder operating in quadrature mode, this counter can count an interval exceeding 250 s before overflow. The incremental encoder *A* and *B* lines are connected to the *A* and

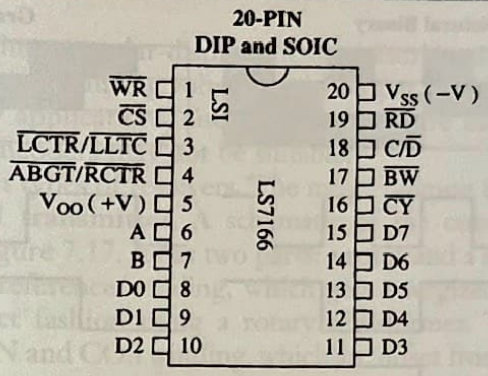


Figure 7.14

Commercial counter IC
(Courtesy of LSI Computer Systems, Melville, NY)

B pins on this IC. This counter uses an 8-bit (pins D0 through D7) three-state I/O bus to communicate with external circuits. An 8-bit bus is used instead of a 24-bit to reduce the number of wires needed but also to be compatible with most microcontrollers/external devices, which have a limited number of I/O lines. An external device can read the counter value by simply sending a preset control value over the I/O bus. This causes the 24-bit counter value to be transferred to the output port of the counter or the output latch. The three byte contents of the output latch are then transferred by performing three successive read operations of the output latch where (after each byte is read) the address pointer for the next byte is automatically incremented.

While incremental encoders do not have the limitations of potentiometers in terms of limited motion range and friction due to contact, they need to be 'homed' before they can be used in a motion-control application. In *homing*, the motor is rotated in one direction until a reference signal changes state. The output of the counter is recorded at this location, and displacements are measured with respect to this reference counter or home position. Most commercial incremental encoders have a third output called the **marker** or **z-channel** that is used in the homing sequence. A limitation of incremental encoders is that homing may not be safe to perform at all times. An example would be a robot arm that uses incremental encoders and is used for operations inside a vehicle frame or in regions with obstacles. If the robot happens to lose power while it is inside the vehicle, the robot will lose its current position information after the power is turned back on. In this case, the robot should not be homed automatically because of the possibility of the robot hitting the vehicle or one of the obstacles. It would be better in this case to use a position sensor that does not need to be homed. Such sensors are called absolute encoders and are discussed in the following section.

7.3.4 Absolute Encoder

An absolute encoder is one which has different track information for different angular positions of the encoder disk. Figure 7.15 shows a layout of a commercial absolute encoder disk. There is no need for homing with an absolute encoder, since each angular position of the disk has a unique output.

Figure 7.15

8-bit commercial absolute encoder disk
(Courtesy of BEI Sensors, Goleta, CA)



Absolute encoders are available with two different types of output: natural binary and gray code. In **natural binary**, the output of the encoder as the disk rotates changes in the normal way that binary numbers increase (i.e., 00, 01, 10, 11, . . .). In **gray code**, the output only changes one bit at time as the disk rotates (i.e., 00, 01, 11, . . .). This makes gray code useful to reduce errors when reading the encoder if all the bits have not changed at the same time. Figure 7.16 shows the disk pattern (shown as linear for ease of display) and the corresponding output of the encoder for a 3-bit natural binary and gray code absolute encoders. Table 7.2

Figure 7.16

Disk pattern and output from each track of an absolute encoder

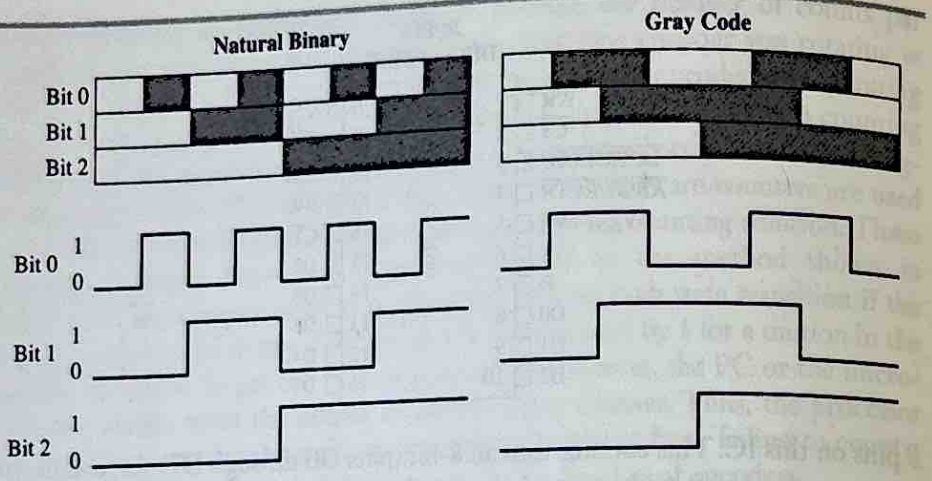


Table 7.2

Encoder output for natural binary and gray code

Position	Angular Segment (degrees)	Encoder Output	
		Natural Binary	Gray Code
1	0–45	000	000
2	45–90	001	001
3	90–135	010	011
4	135–180	011	010
5	180–225	100	110
6	225–270	101	111
7	270–315	110	101
8	315–360	111	100

gives the binary output patterns for both types of encoders for the eight different positions of the encoder disk.

Notice that this 3-bit absolute encoder (which has three tracks) can measure eight distinct absolute positions, each 45 degree in size (i.e., 0–45, 45–90, . . . etc). Commercial absolute encoders have typically 10-bit (or 10 tracks) or higher resolution, which give them an angular resolution of 360/1024 degrees or less. With the use of a geared motor, the resolution of the angular measurement of the output shaft increases by the gear ratio factor.

To use an absolute encoder for **multi-revolution measurement**, multiple disks need to be used. A high-resolution disk is used for the detailed position information and one or more disks are used for counting turns. For example, to use an absolute encoder having a measurement range of 16 revolutions, two disks are used. The second disk will have four tracks (to indicate 16 different turns) and should be coupled to the high-resolution disk through a 16:1 gear ratio. If the primary disk has a 10-bit resolution, then this two-disk encoder will measure 16×1024 or 16384 discrete positions.

Commercial absolute encoders are available with different types of output. These include parallel digital output, which uses a single line for each bit. For a multi-turn encoder with a 14-bit disk, this results in an interface cable that has over 30 wires, which increases the cost of the encoder. A smaller sized cable (and hence lower cost) is obtained if an encoder with SPI output is used. The SPI interface (see Chapter 5) uses only three wires for transmitting the data. Other output formats include DeviceNet™, Profibus, and Interbus.